

Some mechanistic underpinnings of molecular adaptations of SARS-COV-2 spike protein by integrating candidate adaptive polymorphisms with protein dynamics

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Abstract

We integrate evolutionary predictions based on the neutral theory of molecular evolution with protein dynamics to generate mechanistic insight into the molecular adaptations of the SARS-COV-2 Spike (S) protein. With this approach, we first identified Candidate Adaptive Polymorphisms (CAPs) of the SARS-CoV-2 Spike protein and assessed the impact of these CAPs through dynamics analysis. Not only have we found that CAPs frequently overlap with well-known functional sites, but also, using several different dynamics-based metrics, we reveal the critical allosteric interplay between SARS-CoV-2 CAPs and the S protein binding sites with the human ACE2 (hACE2) protein. CAPs interact far differently with the hACE2 binding site residues in the open conformation of the S protein compared to the closed form. In particular, the CAP sites control the dynamics of binding residues in the open state, suggesting an allosteric control of hACE2 binding. We also explored the characteristic mutations of different SARS-CoV-2 strains to find dynamic hallmarks and potential effects of future mutations. Our analyses reveal that Delta strain-specific variants have non-additive (i.e., epistatic) interactions with CAP sites, whereas the less pathogenic Omicron strains have mostly additive mutations. Finally, our dynamics-based analysis suggests that the novel mutations observed in the Omicron strain epistatically interact with the CAP sites to help escape antibody binding.

eLife assessment

This **important** study investigates various variants of the SARS-COV-2 spike protein using established computational methods, complemented by experimental validation efforts. The evidence, bolstered by an evolutionary approach and protein dynamics, is **solid**. Placing this research in the broader context of the field could further enrich the manuscript. It will interest biophysicists focused on allostery and protein evolution.

Introduction

Since 2019, the evolution of SARS-CoV-2 in humans has been characterized by the spread of mutations, many notably found within the Spike (S) glycoprotein. The S protein is directly related to the human immune response to COVID-19 and, as such, has been one of the most studied and targeted proteins in SARS-CoV-2 research (Shang et al. 2020 [DOI](#); Harvey, Carabelli, Jackson, Gupta, Thomson, Harrison, Ludden, Reeve, Rambaut, Consortium, et al. 2021 [DOI](#); Jackson et al. 2022 [DOI](#); Carabelli et al. 2023 [DOI](#); Markov et al. 2023 [DOI](#)). Subsequently, research into the biophysical properties and mutational patterns associated with S protein evolution not only remains critical to understanding the pandemic but also emerges as a useful system to understand the mechanics of molecular adaptation within viruses.

For successful infection of a human host, the S glycoprotein of SARS-CoV-2 binds to the human ACE2 (hACE2) receptor through its receptor-binding domain (RBD). Evidence indicates that fine-tuning S protein interactions with hACE2 significantly affects viral reproduction (Rehman et al. 2020; Saputri et al. 2020 [DOI](#); Rochman et al. 2021 [DOI](#)). Previous evolutionary studies show a complex network of interactions among mutated residues (Changeux and Edelstein 2005 [DOI](#); Doshi et al. 2016 [DOI](#); O'Rourke et al. 2016 [DOI](#); Mishra and Jernigan 2018 [DOI](#)). Therefore, there has been a vast effort to uncover which mutations are important steps of adaptation for the S protein (Cagliani et al. 2020 [DOI](#); Damas et al. 2020 [DOI](#); Singh and Yi 2021 [DOI](#); Kistler et al. 2022 [DOI](#); Maher et al. 2022 [DOI](#); Neher 2022 [DOI](#)). In particular, a significant aspect of many such studies was a focus on understanding adaptive mutations of SARS-CoV-2 that contributed to the leap to human hosts (Cagliani et al. 2020 [DOI](#); Damas et al. 2020 [DOI](#); Singh and Yi 2021 [DOI](#); Starr, Zepeda, et al. 2022). This is because SARS-CoV-2 has continuously mutated since its early detection (Kistler et al. 2022 [DOI](#)), causing the emergence of CDC-designated “variants of concern” (VOCs) that may be driven by an accelerated substitution rate may (Tay et al. 2022 [DOI](#)).

Predicting how new mutations impact the biophysical properties of the S protein remains a challenge, let alone explaining their complex interactions with one another and how they might affect hACE2 binding; many factors affect hACE2 interactions. Binding affinity with hACE2 can be enhanced directly through stronger receptor interactions or mediated through changes in RBD opening (Natália Teruel et al. 2021a; Zhang et al. 2021 [DOI](#); Díaz-Salinas et al. 2022 [DOI](#)). The RBD exhibits both ‘closed’ and ‘open’ conformational states. In the closed state, the RBD is shielded from receptor binding. In the open state, the RBD is accessible for hACE2 binding (Kirchdoerfer et al. 2016; Gur et al. 2020 [DOI](#); Henderson et al. 2020 [DOI](#); Hoffmann et al. 2020a [DOI](#)). While some mutations may affect the transition between these states (Henderson et al. 2020 [DOI](#); Yurkovetskiy et al. 2020 [DOI](#); Gobeil, Janowska, McDowell, Mansouri, Parks, Manne, et al. 2021; Sztain et al. 2021 [DOI](#); Zhang et al. 2021 [DOI](#); Shoemark et al. 2022 [DOI](#)), other mutations may allosterically regulate RBD openings through Furin cleavage site (residue ID range: 681-695) interactions to regulate hACE2 binding (Deng et al. 2021 [DOI](#); Gobeil, Janowska, McDowell, Mansouri, Parks, Stalls, et al. 2021; Khan et al. 2021 [DOI](#); Laiton-Donato et al. 2021 [DOI](#)).

Moreover, as new mutations accumulate, culminating in the emergence of a new VOC, these mutations must occur on varied sequence backgrounds containing neutral, nearly-neutral, and adaptive mutations. While many studies have explored the impacts of individual mutations, VOCs result in a substantial difference in protein function as compared to their individual effects (Moulana et al. 2022a [DOI](#); Starr, Greaney, Hannon, et al. 2022; Moulana et al. 2023 [DOI](#); Witte et al. 2023 [DOI](#)). Here we integrate an evolutionary approach with protein dynamics analysis to address the fundamental mechanisms of mutations dictating VOCs and the impact of their epistatic interaction on the function of the S protein. Many earlier studies have combined phylogeny and evolutionary theory to identify adaptive mutations and analyze how the viral sequence has changed over time (Frost et al. 2018 [DOI](#); Boni et al. 2020 [DOI](#); Cagliani et al. 2020 [DOI](#); Damas et al.

2020 [Tang et al. 2020](#)). Similarly, we first use a well-established Evolutionary Probability (EP) approach (Liu et al. 2016 [Liu et al. 2016](#)) that utilizes phylogenetic trees in combination with the neutral theory of molecular evolution to determine Candidate Adaptive Polymorphisms (CAPs) using the early Wuhan sequence as a variant. CAPs are substitutions in SARS-CoV-2 that are rarely observed in other closely related sequences (**Figure 1A** [Liu et al. 2016](#)), which implies a degree of functional importance and makes them candidates for adaptation (Liu et al. 2016 [Liu et al. 2016](#)). Adaptation in this case means a virus which can successfully infect human hosts. As CAPs are unexpected polymorphisms under neutral theory, their existence implies a non-neutral effect. This can come in the form of functional changes (Liu et al. 2016 [Liu et al. 2016](#)) or compensation for functional changes (Ose et al. 2022 [Ose et al. 2022](#)). Therefore, we suspect that these CAPs may be partially responsible for the functional change allowing the infection of human hosts. In support of this method, we find an overlap between our list of sites containing CAPs and putative adaptive sites identified by others (Cagliani et al. 2020 [Cagliani et al. 2020](#); Singh and Yi 2021 [Singh and Yi 2021](#); Kistler et al. 2022 [Kistler et al. 2022](#); Starr, Zepeda, et al. 2022). Second, we use a suite of computational tools to analyze how CAPs that arose in the early and late phases of the COVID-19 pandemic modulate the dynamics of the S protein. We also explore the complex interactions between these sets of CAPs to gain mechanistic insight into the behavior of molecular adaptation involving the S protein. In particular, we focused on how mutations modulate protein dynamics, as we and others have previously found that rather than changing a protein's structure solely, mutations modulate conformational dynamics leading to changes in biophysical properties such as stability, flexibility, and allosteric dynamic coupling, any of which may affect protein binding (Swint-Kruse et al. 1998 [Swint-Kruse et al. 1998](#); Keskin et al. 2000 [Keskin et al. 2000](#); Bhabha et al. 2013 [Bhabha et al. 2013](#); Nussinov, R., Tsai, C.-J 2013; Campbell, E. et al. 2016; Ma and Nussinov 2016 [Ma and Nussinov 2016](#); 201; Saavedra et al. 2018 [Saavedra et al. 2018](#); Kuzmanic et al. 2020 [Kuzmanic et al. 2020](#)).

With this evolutionary-dynamics unified approach, we aim to answer the following questions about VOCs: Are all the mutations in VOCs adaptive in nature? Are they coupled to one another or provide some measure of biophysical, dynamical, or mechanical epistasis? While many of these mutations are found within the RBD domain, numerous others are located distal to this region; hence, we aim to investigate the functional roles of distal mutations, particularly from a protein dynamics perspective. Our integrated analysis revealed that protein dynamics play a significant role in the evolution of the S protein. The flexibility of sites within the S protein shows a strong, direct correlation with substitution rate, and newly evolving CAPs are mostly additive mutations that modulate the dynamics of the hACE2 binding site. Yet other CAPs, 346R, 486F, and 498Q, show highly epistatic (i.e., non-additive) modulation of the hACE2 binding site and provide immune escape benefits.

Results and Discussions

Candidate Adaptive mutations in the Spike protein are recognized via Evolutionary Probabilities

SARS-CoV-2 is part of a family of coronaviruses, many of which infect mainly animals and are less capable of infecting humans (Dicken et al. 2021 [Dicken et al. 2021](#)). Therefore, to identify the most likely mutations responsible for the infection of human hosts (i.e., putative adaptive mutations for humans), we estimated the (neutral) evolutionary probability (EP) scores of mutations found within the S protein (Liu et al. 2016 [Liu et al. 2016](#)). EP scores of the amino acid variants of the S protein were obtained using a Maximum Likelihood phylogeny (Kumar et al. 2018 [Kumar et al. 2018](#)) built from 19 orthologous coronavirus sequences. Sequences were selected by examining available non-human sequences with a sequence identity of 70% or above to the human SARS CoV-2's S protein sequence. This cutoff allows for divergence over evolutionary history such that each amino acid position had ample time to experience purifying selection, whilst limiting ourselves to closely related coronaviruses. (**Figure 1A** [Liu et al. 2016](#)). The likelihood of finding a particular amino acid in the sequence is then determined using a Bayesian framework, with calculations carried out by MEGA X software (Kumar et al. 2018 [Kumar et al. 2018](#)). As apparent in the name, EP scores obtained for the amino acids in the

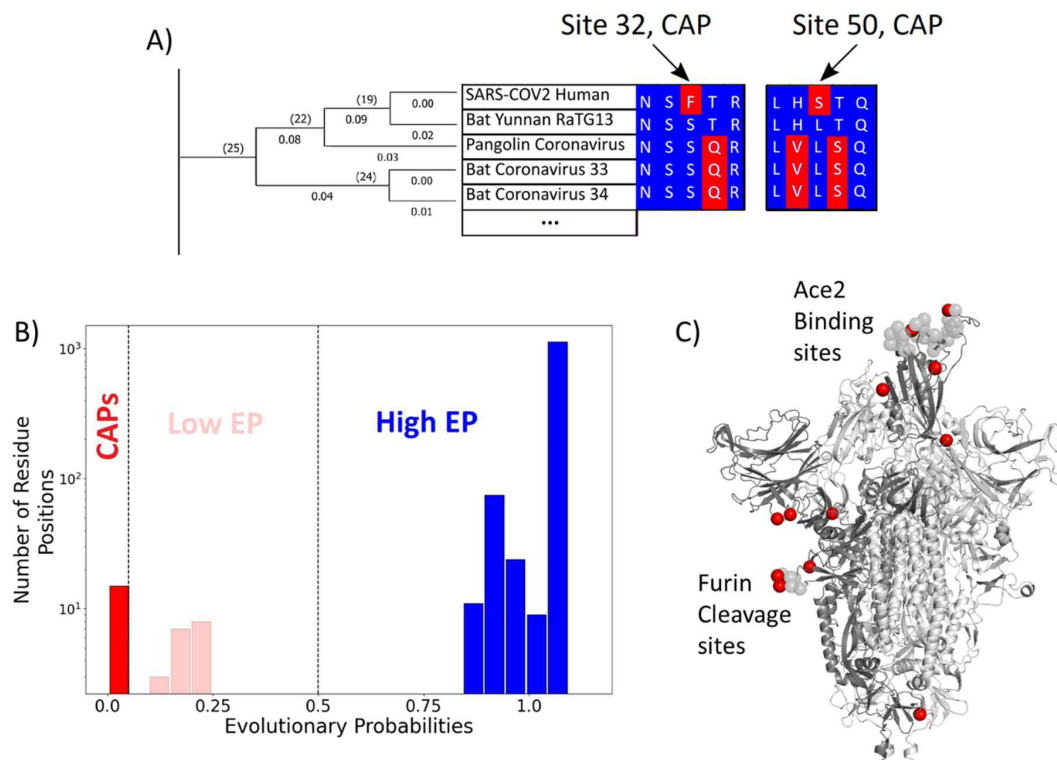


Figure 1.

(A) The evolutionary probabilities (EP) of each amino acid in the S protein sequence are calculated by taking the multiple sequence alignment of the S proteins through their evolutionary tree and using Bayesian inferences to determine the likelihood of finding a particular residue at a particular location within a given sequence. Simply, if the residue is found at a location 'x' in closely related sequences, it will have a higher EP at location 'x' in the target sequence. Residues with an EP <0.05 in the target sequence are CAPs (Red). CAPs are found at sites 32, 50, 218, 346, 372, 478, 486, 498, 519, 604, 681, 682, 683, 684, and 1125. (B) The distribution of EP scores of the wild-type residues in the S protein. Here, lower EP scores are shown in red, and higher EP scores in blue. While the vast majority of the wild-type (reference) protein consists of high EP residues, a few residues have low EP. (C) The CAPs are also highlighted as red spheres in the open configuration of the S protein, with the open chain in a darker shade. We observe that a majority of the CAP positions reside at the receptor binding domain (RBD) and the Furin cleavage site (676-689; (Wrobel et al. 2020)) shown as transparent light gray spheres.

sequence provide information regarding the likelihood of finding them at their position, given the history of the sequence. Amino acid residues receiving low EP scores (<0.05) at a position are less likely to be found in a given position within the sequence because they are non-neutral. Generally, positions with low EP amino acids are far less common than those containing mutations with high EP, a trend also realized in the CoV-2 S protein (**Figure 1B**).

Of particular interest is an observed evolutionary change where an amino acid with high EP is replaced by an amino acid residue with low EP. While amino acids with low EP should be harmful or deleterious to viral fitness due to functional disruption or change, fixation of a low EP amino acid at a position suggests an underlying mechanism for natural selection to operate. These fixed, low EP mutations are called candidate adaptive polymorphisms (CAPs) as they are predicted to alter protein function, and adaptive pressures may drive their prevalence (Patel et al. 2018). Indeed, there is an overlap between these CAPs and the mutations suggested by other methods to be adaptive for the S protein (Cagliani et al. 2020; Singh and Yi 2021; Kistler et al. 2022; Starr, Zepeda, et al. 2022). Within these studies, sites 478, 486, 498, and 681 have been implicated in SARS-CoV-2 evolution, leaving the remaining 11 CAPs as undiscovered candidate sites for adaptation.

Interestingly, most of the CAP residues are at functionally critical sites, including the receptor binding domain (RBD) and the Furin cleavage site (**Figure 1C**). As mentioned earlier, the RBD plays a key role in initiating the infection of a healthy cell by binding with the host organism's ACE2 protein. Before ACE2 binding, one chain of the homotrimer comprising the S protein must open to expose the RBD (Kirchdoerfer et al. 2016; Henderson et al. 2020; Hoffmann et al. 2020a; Sztain et al. 2021). The Furin cleavage site plays a key role in the opening process as the binding of host cell protease Furin aids in the cleavage of the S protein into two domains: S1 and S2 (Wrobel et al. 2020: 13). Another host cell protease, TMPRSS2, facilitates viral attachment to the surface of target cells upon binding either to sites Arg815/Ser816, or Arg685/Ser686 which overlap with the furin cleavage site 676-689, further emphasizing the importance of this area (Hoffmann et al. 2020b; Fraser et al. 2022). Similar cleavage sites have been found in related coronaviruses, including HKU1 and Middle East respiratory syndrome coronavirus (MERS-CoV), which infects humans (Chan et al. 2008: 1; Millet and Whittaker 2014; Millet and Whittaker 2015), and the acquisition of similar cleavage sites is associated with increased pathogenicity in other viruses such as the influenza virus (Steinhauer 1999). Interestingly, however, CAPs do not display such an overwhelming tendency to occur at well-known critical sites within human proteins studied with similar methods (Ose, Campitelli, et al. 2022), yet mutations at those sites are associated with disease, indicating their critical role in inducing functional change. Therefore, the identified CAPs in the S protein, which are signs of recent evolution, can provide mechanistic insights regarding the molecular adaptation of the virus. In particular, we aimed to analyze how these CAP positions in the S protein modulate the interaction with hACE2 using our protein dynamics-based analysis (Gerek and Ozkan 2011; Nevin Gerek, Z., Kumar, S., Banu Ozkan, S. 2013; Larrimore et al. 2017; Kumar, A., Glembo, T.J., Ozkan, S.B. 2015b).

Asymmetry in communications among the network of interactions in Spike describes how CAPs regulate the dynamics of the Spike protein

A mutation at a given amino acid position inevitably not only alters local interactions, but this change cascades through the residue-residue interaction network, which gives rise to a variation in native ensemble dynamics to modulate function (Dror et al. 2012; Labbadia and Morimoto 2015; Sekhar and Kay 2019; Campitelli et al.). Many groups have already examined the conformational dynamics of the spike protein using normal mode analysis to explore mutation sites and interactions with different receptors (Zhou et al. 2020; Majumder et al. 2021; Natália Teruel et al. 2021b; Natalia Teruel et al. 2021; Verkhivker 2022). However, our study will focus mainly on the role of CAPs, of which F486 and Q498 have already been identified through

Perturbation Response Scanning (PRS) as potential allosteric sites (Verkhivker 2022 [↗](#)). Thus, we analyze the internal dynamics of the system to understand the functional role of CAPs in S proteins. This analysis allows us to gain a mechanistic understanding of the relationship between CAP mutations and biophysical outcomes (Natália Teruel et al. 2021a). First, we implement the Dynamic Coupling Index (DCI) approach to study long-distance coupling between the CAPs and the hACE2 binding sites emerging from the 3D network of interactions across the S protein system. DCI calculation combines Perturbation Response Scanning and Linear Response Theory to capture the strength of a displacement response for position i upon perturbation of position j , relative to the average fluctuation response of position i to all other positions in the protein. It represents the strength of dynamic coupling between positions i and j upon perturbation to j (Larrimore et al. 2017 [↗](#); Kumar, A., Glembo, T.J., Ozkan, S.B. 2015b).

Further, asymmetry can be captured in the DCI values, as dynamic coupling is not necessarily symmetric due to an anisotropic interaction network. That is, each amino acid has a set of positions to which it is highly coupled, and this anisotropy in connections gives rise to unique differences in coupling between a given i, j pair of amino acids which do not have direct interactions. By calculating the coupling of the hACE2 binding interface in the RBD with respect to the CAP residue positions and vice versa, we can generate DCI_{asym} (Figure 2A [↗](#)) as the difference between the normalized displacement response of position j upon a perturbation to position i (DCI_{ij}) and the normalized displacement response of position i upon a perturbation to position j (DCI_{ji}) (See Methods). If the DCI_{asym} values significantly differ from zero, it shows asymmetry in coupling and presents a cause-effect relationship between the i, j pair in terms of force/signal propagation. This metric has been used previously in a variety of systems to analyze the unique behavior of positions within a protein and a given position's propensity to effect biophysical changes upon mutation, particularly at long distances (Modi and Ozkan 2018 [↗](#); Campitelli and Ozkan 2020 [↗](#); Kolbaba-Kartchner et al. 2021 [↗](#); Ose, Butler, et al. 2022; Kazan et al. 2023 [↗](#); Campitelli et al. 2020b [↗](#)).

Recent work from our group has shown an enhancement in cross-chain communication within the main protease of SARS COV-2 compared to SARS COV-1 (Campitelli et al. 2022 [↗](#)). Furthermore, previous studies have shown that allosteric inter-chain communication is important to S protein function (Zhou et al. 2020 [↗](#); Spinello et al. 2021 [↗](#); Tan et al. 2022 [↗](#); Xue et al. 2022 [↗](#)). In support of these findings, we observe through DCI_{asym} that when the S protein is in its pre-fusion conformation with one chain open, the CAPs in the closed chains have negative coupling asymmetry with respect to the hACE2 binding site interface in the RBD-open chain. This indicates an allosteric control where the hACE2 binding site is dominated by the dynamics of the CAPs in closed chains (Figure 2B [↗](#), yellow bars). As this open state RBD is critical for the viral infection of host cells (Kirchdoerfer et al. 2016), our results suggest that this type of closed-to-open cross-chain interaction is important for viral proliferation. Our prior studies on DCI_{asym} show a similar trend in Lactose Inhibitor (LacI), a protein with a functional role in gene expression through binding DNA. The allosteric mutations (i.e., mutations on the sites that are far from the DNA binding sites) that alter DNA binding affinity not only exhibited unique asymmetry profiles with the DNA binding sites of LacI, but also regulated the dynamics of these binding sites (Campitelli et al. 2020b [↗](#)).

Similarly, it is possible that mutations to such residue positions within the S protein can be used to regulate the dynamics of the hACE2 bindings sites of the open RBD state. We, therefore, propose that the residue positions with CAP substitutions hold the potential for mutations in the spike sequence which can alter the opening and closing dynamics of the RBD domain. This hypothesis is further supported by many mutations already observed at these residue positions which alter the infection rate (Brister et al. 2015 [↗](#)). Interestingly, residues responsible for extremely low asymmetry values (< -4) lie overwhelmingly in the region 476–486. These same residues were suggested to stabilize S protein dynamics and prime it for host Furin proteolysis (Raghuvamsi et al. 2021 [↗](#)).

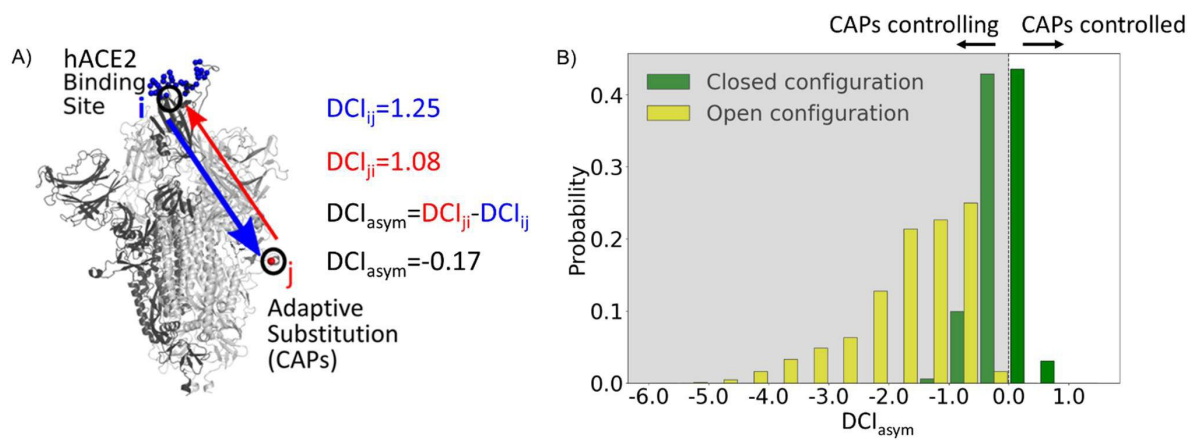


Figure 2.

(A) Schematic representation of DCI_{asy} . (B) DCI_{asy} of CAP residue positions with the binding interface of RBD in the open chain. Residues in the closed chains with a low EP amino acid in the reference sequence dominate the binding site interface of RBD in the open chain. There is a significant difference between the asymmetry profiles of the closed ($M = -0.06$, $SD = 0.33$) and open ($M = -1.68$, $SD = 0.89$) conformations ($p < .001$).

Moreover, as a control, we performed the same analysis on the S protein with the RBD domains of all chains in the closed configuration. In this case, we observed that the DCI_{asym} of the CAPs residue positions with respect to the hACE2 interface in the other chains yields a largely symmetric distribution about 0 (**Figure 2B**, green bars). This verifies that the asymmetry in the coupling of CAPs with the exposed binding site interface in pre-fusion configuration results from one of the RBDs opening up and further suggests the allosteric role played by CAPs in locking the S protein in the RBD open state.

Dynamics analysis shows that rigid sites tend to be more highly conserved than flexible sites

CAPs represent important S protein amino acid changes between related coronaviruses across multiple species and the Wuhan-Hu-1 reference sequence (MN908947). Since SARS-CoV-2 first spread to humans, it has continued to mutate and evolve rapidly, particularly regarding the S protein (Amicone et al. 2022; Liu et al. 2022; Tay et al. 2022). Just as mutations leading to the Wuhan strain caused an increase in binding affinity to hACE2, continued evolution in human hosts has resulted in further altered binding affinities as well as different phenotypic outcomes for those infected (Ali et al. 2021; Barton et al. 2021; Ozono et al. 2021).

We explore whether protein dynamics has played a role in the selection of mutational sites during the evolution of the S protein since 2019. Our previous work has indicated that rate of evolution per positional site exhibits a positive correlation with positional flexibility; generally, positions that exhibit higher flexibility are also sites that experience a higher number of amino acid substitutions (Liu and Bahar 2012; Maguid et al. 2008; Maguid et al. 2006; Mikulska-Ruminska et al. 2019; Nevin Gerek, Z., Kumar, S., Banu Ozkan, S. 2013). To confirm these findings for the evolution of the S protein using the sequenced variants of infected humans, we examine the flexibility of the S protein, an analysis conducted in other studies which resulted in several important findings (Nguyen et al. 2020; Socher et al. 2021; Natália Teruel et al. 2021b; Natalia Teruel et al. 2021; Pipitò et al. 2022; Verkhivker 2022; Abduljalil et al. 2023). For example, Teruel et al. (2021) used their Elastic Network Contact Model to find how certain highly observed mutations make the open state more rigid and the closed state more flexible. For our own flexibility analysis, we measure the site-specific amino acid flexibility using the Dynamic Flexibility Index (DFI). Using the same mathematical foundation as DCI, DFI evaluates each position's displacement response to random force perturbations at other locations in the protein (Gerek and Ozkan 2011; Nevin Gerek, Z., Kumar, S., Banu Ozkan, S. 2013), and it can be considered a measure of a given position's ability to explore its local conformational space. We found that the covid-19 S protein shows the expected high correlation between the occurrence of mutations and site flexibility (**Figure 3**) when we compare %DFI (DFI ranked by percentile) to the average number of variants per position found within a given %DFI bin. Previous studies have indicated that rigid residues are critical for functional dynamics, thus more likely to impact function if mutated and, generally, can lead to a loss of function and thus more conserved (Kim, H. et al. 2015; Butler et al. 2018; Modi, Risso, et al. 2021; Modi, Campitelli, et al. 2021; Kazan et al. 2022; Ose, Butler, et al. 2022; Stevens et al. 2022; Campitelli et al. 2020a; Kumar, A., Glembo, T.J., Ozkan, S.B. 2015b). This analysis also agrees with these previous studies and highlights the power of negative selection, in line with the neutral theory of molecular evolution, stating that deleterious mutations (i.e., those on the rigid positions) should be eliminated and therefore not observed (Kimura, Motoo 1983).

Continued mutations within human hosts have resulted in a multitude of variants. Indeed, by fitting various molecular clock models to genome sequence data, VOC emergence is punctuated by an episodic period of rapid evolution, with a substitution rate of up to 4-fold greater than the background substitution rate (Kumar et al. 2021; Tay et al. 2022). With such an aggressive

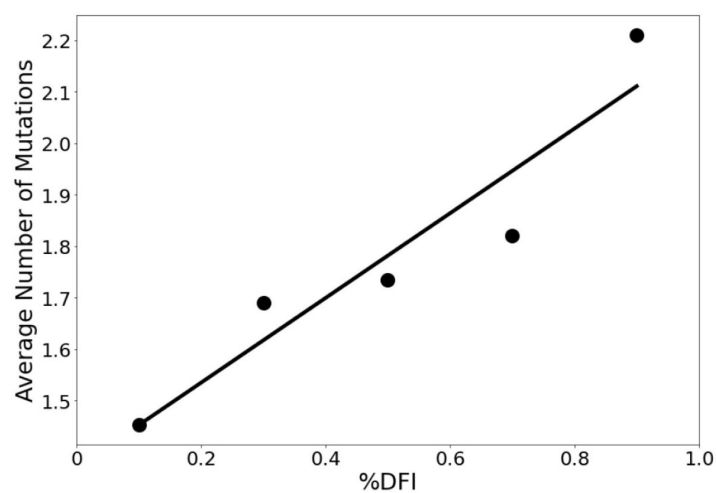


Figure 3.

The average number of variants observed among residues of different flexibility.

Residues were sorted into one of five bins based on flexibility. After that, the average number of variants for residues within that bin was calculated. Here, the number of variants is defined as the number of different amino acid varieties found at that site. Mutational data was calculated across approximately 24,000 SARS-CoV-2 S protein sequences from the NCBI Datasets Project (Brister et al. 2015 [\[link\]](#)). Residue flexibility, as reported here via %DFI, was computed using PDB id 6vsb from the Protein DataBank (Berman, H.M. et al. 2000). More rigid residues tend to have fewer variants. ($r = 0.94$).

evolutionary rate, we are finding VOCs to consist of a number of different characteristic mutations, almost all of which are CAPs. Sequence differences between the various VOCs used in this paper can be found in **Supplementary Figure S1** [↗](#).

To explore the dynamic effects of the evolution of the Spike in humans, we examine asymmetry with these new potentially adaptive sites, namely the low EP (CAP) characteristic mutation sites observed in the Delta variant, the widely dominant variant from December 2021 to January 2022 (Thye et al. 2021 [↗](#)), and the Omicron variant, a highly transmissible variant whose lineages have remained dominant since January 2022 (Kim et al. 2021) (**Figure 4** [↗](#)). This analysis revealed a mechanism similar to that for the CAPs in the reference protein (**Figure 2** [↗](#)), as the open-chain binding interface is also allosterically controlled by these potentially new adaptive sites. Regarding this, we see that the asymmetry is much more pronounced in observed mutations of Omicron variants suggesting that these new mutations have a stronger power in controlling the dynamics of open chain hACE2 binding interface compared to those observed in Delta variants. We surmise that the difference in virulence and infection rates between Omicron and Delta (Earnest et al. 2022; Bager et al. 2021; Sheikh et al. 2021; Twohig et al. 2022; Houhamdi et al. 2022 [↗](#); Menni et al. 2022 [↗](#)) might be due to these specific CAPs within each variant and their differences in allosterically controlling the dynamics of open RBD binding sites as also observed in the DCI_{asym} analysis of the Wuhan variant in **Figure 2** [↗](#).

Experimental results motivate the use of EpiScore within the SARS-CoV-2 Spike protein

The fact that the identified CAPs in the reference protein and the more recently evolved CAPs of Delta and Omicron variants both show a high degree of control over the functional sites begs the question: what is the complex interaction between these previous and new CAP sites? Motivated by this concept, we explore the interplay of mutational pairs to understand the effects of the specific amino acid backgrounds associated with these two predominant variants. Some CAP sites in Delta and Omicron have already been considered adaptive (Kemp et al. 2021 [↗](#); Kistler et al. 2022 [↗](#); Maher et al. 2022 [↗](#); Neher 2022 [↗](#)).

It is well understood that the impact of even a single mutation to a protein sequence can sometimes dramatically alter the biophysical behavior of the system. However, the mechanistic impact of point mutations can only be fully understood when the sequence background upon which it is made is accounted for. This means that, in the case of strains with multiple mutations, the interplay between mutated positions will ultimately impact a protein as an aggregate behavior, where the presence of previous mutations may strongly (or weakly) influence some mutations. This concept of non-additivity is known as epistasis. In fact, studies of evolutionary pathways of mutations have suggested that a majority of the mutations have a second or a higher order epistasis among them (Bershtein et al. 2006 [↗](#)). Nature exploits this higher order complex relationship between the mutations to evolve their function.

To computationally capture and interpret the pairwise effects of mutations, we have developed an in-house computational tool called EpiScore (**Figure 5A** [↗](#)). Here, we evaluate how a given position pair i, j may affect other critical positions k of the protein. EpiScore is the relative coupling strength to a position k when positions i and j are perturbed *simultaneously* compared to the average dynamic coupling strength of i to k and j to k . EpiScore has previously been used successfully to capture overarching trends in GB1 deep mutational scan data as well as specific instances of the development of antibiotic resistance in various enzymatic systems (Campitelli and Ozkan 2020 [↗](#)). An EpiScore of 1 indicates perfect coupling additivity, and deviations from this value represent non-additive behavior between position pairs and functionally important sites. Prior EpiScore work has shown a difference in EpiScore between the sites of compensatory and

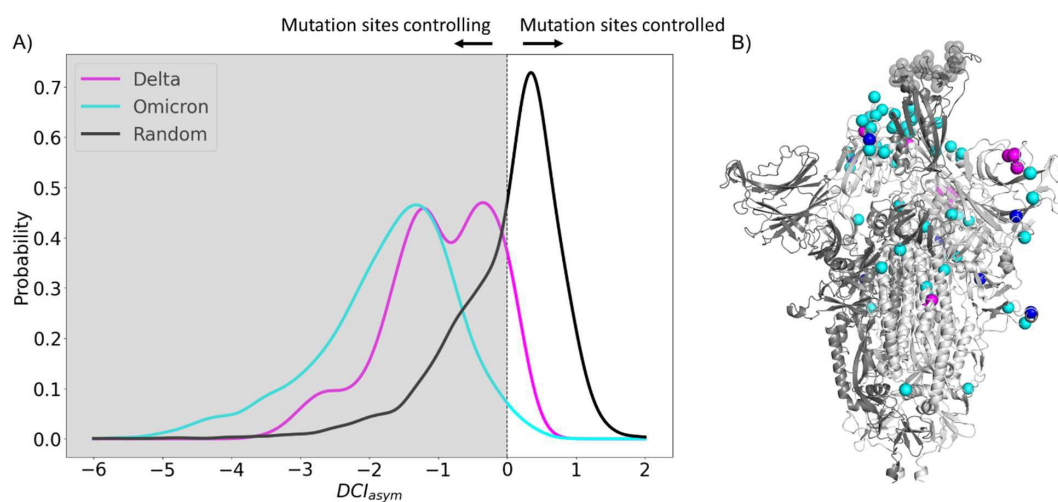


Figure 4.

(A) DCI_{asym} with low EP characteristic mutation sites of Delta or Omicron strains in the closed chains and the binding interface of RBD in the open chain. Delta displays a second peak closer to zero, suggesting that Delta mutation sites ($M = -0.98$, $SD = 0.80$) have less allosteric control over the hACE2 binding sites than Omicron mutation sites ($M = -1.74$, $SD = 1.00$) ($p < .001$). However, both sets of sites have far more control over hACE2 binding sites than expected, based on a random control group ($M = 0.03$, $SD = 0.85$) ($p < .001$). (B) S protein structure showing binding interface sites (transparent gray), Delta mutation sites (magenta), Omicron mutation sites (cyan), and sites mutated in both Omicron and Delta (Blue).

non-compensatory mutations, where both yield distributions with peaks around 1, but non-compensatory mutations show higher deviation in their EpiScore distribution (Ose, Campitelli, et al. 2022 [\[1\]](#)).

Many studies have confirmed epistasis between residues within the S protein (Moulana et al. 2022a [\[2\]](#); Starr, Greaney, Hannon, et al. 2022; Moulana et al. 2023 [\[3\]](#); Witte et al. 2023 [\[4\]](#)). These epistatic residues can have various effects on hACE2 or antibody binding. Within SARS-CoV-2, here we calculate the EpiScore (**Figure 5B** [\[5\]](#)) of a set of mutation pairs used by Moulana et al. (Nature comm 2022) and compare our results to quantified epistatic effects determined by the experimental hACE2 binding affinity of “first-order” single mutation variants compared to “second-order” mutation pairs. Our EpiScore results and the experimentally determined epistasis both captured highly epistatic behavior among residues 493, 496, 498, 501, and 505, as well as a lack of epistatic behavior for residues 339, 371, 373, and 375, however EpiScore generally showed higher epistasis values than experiment for residues 417, 440, 446, 477, 478, and 484.

Episcore highlights the epistatic relationship between the recent adaptive mutations in VOCs and the CAPs of the Wuhan reference

Seeking further to understand the role of epistasis within S protein variants, we explored the possibility of epistatic relationships between the CAPs of the Wuhan variant and the new CAPs in VOCs. Thus, we computed the EpiScore of these CAP positions in the closed RBD domains (i.e., chain B and C) with respect to functional hACE2 binding interface sites of the open RBD domain chain (chain A) (**Figure 5B** [\[5\]](#)) and obtained EpiScore distributions.

To contrast these variants, Omicron (**Figure 6** [\[6\]](#), cyan) shows a high proportion of additive mutations compared to the Delta variant (**Figure 6** [\[6\]](#), magenta), with a peak centered on 1. The comparatively more pathogenic Delta variant exhibited many non-additive mutations with EpiScores below one. This again suggests that the cross-communication between the open and closed chain of the S protein is important for regulating the function. Four out of seven low EP Delta mutation sites used in this analysis often resulted in EpiScores below 1. Each of those is found in the N-terminal domain (NTD) on or near the N3 loop and is implicated in antibody escape in recent studies (Chi et al. 2020 [\[7\]](#); Weisblum et al. 2020 [\[8\]](#); Harvey, Carabelli, Jackson, Gupta, Thomson, Harrison, Ludden, Reeve, Rambaut, COVID-19 Genomics UK (COG-UK) Consortium, et al. 2021 [\[9\]](#); Klinakis et al. 2021 [\[10\]](#); Cantoni et al. 2022 [\[11\]](#)). The low EpiScores of NTD mutations suggest that they dampen the control of Wuhan variant CAPs over the hACE2 binding sites in addition to their effects on antibody binding. It is possible that what the Delta variant gained in transmission rate also came with being more harmfully pathogenic due in part to negatively epistatic interactions. In contrast, the mutations leading to the development of the Omicron strain were additive with respect to Wuhan variant CAPs, possibly leading to a lower pathogenicity and higher effective immune escape, resulting in an overall higher transmission rate. Another possible explanation for the higher transmission rate of Omicron comes from a different normal mode analysis study, which found that despite reduced binding affinity with hACE2, Omicron showed increased occupancy of the open state compared to the closed state, which would increase chances of interaction with hACE2 (Natalia Teruel et al. 2021). It is worth noting that other variants contain NTD mutations which result in low EpiScores, however the proportion of these mutations within the set is considerably less than in Delta (**Supplementary Figure S3** [\[12\]](#)).

One of the more notable features of generated EpiScore distributions is the presence of a tail of values upward of 2.0, indicating highly epistatic behavior. Interestingly, these tails are largely due to three different CAPs: 346R, 486F, and 498Q. Those residues are nearby one another within the protein structure and have been reported to play a role in antibody binding, either being known antibody binding sites (346R and 486F) or having received very high antibody accessibility scores (498Q) (Harvey, Carabelli, Jackson, Gupta, Thomson, Harrison, Ludden, Reeve, Rambaut, COVID-19 Genomics UK (COG-UK) Consortium, et al. 2021; Raghuvamsi et al. 2021 [\[13\]](#)). These observed high

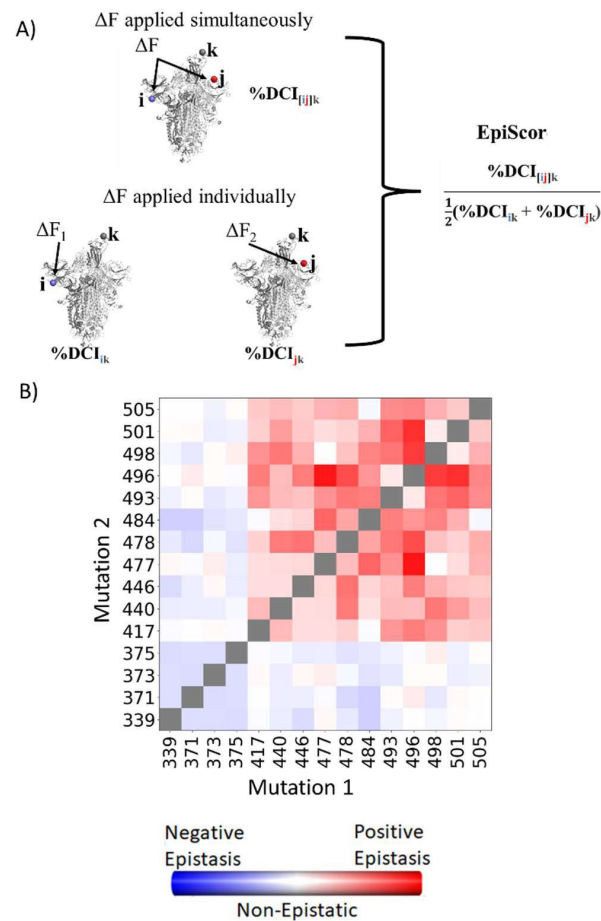


Figure 5.

(A) Schematic representation of cross-chain EpiScore, describing i , j , in chain B and C respectively and its impact in RBD binding position k in the open RBD conformer chain A. (B) Colors indicate EpiScore values for given mutation pairs, averaged over hACE2 binding sites. Cross-chain residue pairs in the upper right tend to be highly epistatic, similar to pairwise second-order interaction coefficients from Moulana et al. (Nature comm 2022).

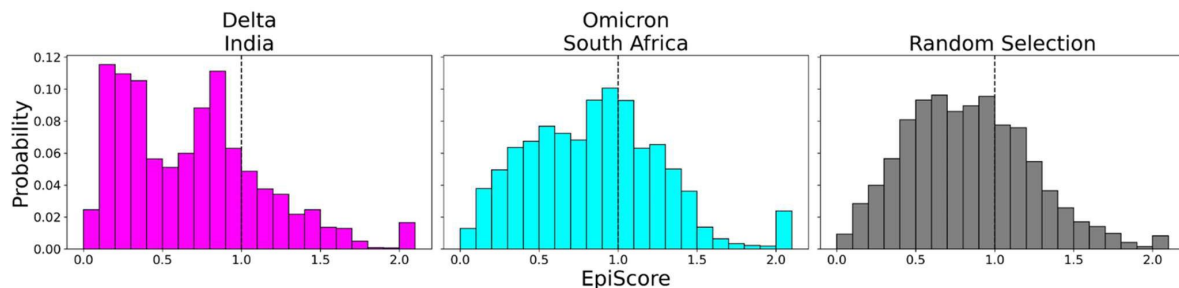


Figure 6.

EpiScores with i = low EP Delta mutation sites (magenta), low EP Omicron mutation sites (Cyan), and a random selection of sites (gray), j = low EP sites in the Wuhan variant, and k = the binding interface of the open chain. EpiScores using sites of either variant (Delta: $M = 0.70$, $SD = 0.50$, Omicron $M = 0.86$, $SD = 0.46$) are significantly different ($p < .001$) from a set of EpiScores using random sites ($M = 0.84$, $SD = 0.41$), but the distribution for Delta variants differs much more from the other two. EpiScores for other variants can be found in **Supplementary Figure S2**.

EpiScore values also support other studies indicating the epistatic interactions between these CAPs and the mutations of the VOCs within the S protein are crucial for maintaining binding affinity of hACE2 whilst evading immunity (Hong et al. 2022 [↗](#); Moulana et al. 2022b [↗](#); Starr, Greaney, Stewart, et al. 2022).

Inspection of EpiScores of Delta and Omicron potentially adaptive mutation sites with only CAP site 486F (a binding site for both hACE2 and antibodies) (Huang et al. 2020 [↗](#); Ali et al. 2021 [↗](#); Harvey, Carabelli, Jackson, Gupta, Thomson, Harrison, Ludden, Reeve, Rambaut, COVID-19 Genomics UK (COG-UK) Consortium, et al. 2021; Raghuvamsi et al. 2021 [↗](#)) shows highly epistatic interactions at other hACE2 binding sites (**Figure 6** [↗](#)). However, within a recent and rapidly spreading subvariant of Omicron, XBB 1.5, we see a mutation of S to P, a rare double nucleotide mutation, at site 486 (preceeding Omicron variants included mutation F486S from the original CAP). This new variant has unprecedented immune escape capabilities, resisting neutralizing antibodies almost entirely (Qu et al. 2023 [↗](#)). EpiScores of other XBB 1.5 specific mutation sites with 486P are almost entirely greater than 1, showing an even higher degree of epistasis with the binding sites of RBD (**Figure 7** [↗](#)). These results present a threefold importance for the S486P mutation: Not only does this residue alter antibody (i.e., immune escape) and hACE2 binding by directly modifying a binding site, but it may also be responsible for modifying hACE2 binding via epistatic cooperation with other co-occurring mutations.

Change in flexibility of RBD binding site correlates with experimental binding affinities for Omicron and Omicron XBB variants

Experimental studies have tracked hACE2 binding for different variants since the virus first spread (Ali et al. 2021 [↗](#); Barton et al. 2021 [↗](#); Ozono et al. 2021 [↗](#); Wu et al. 2022 [↗](#)). Within the Omicron variant, for example, characteristic mutations on the RBD are shown to increase the overall binding affinity of the virus to the ACE2 receptor, which is suspected to allow it to spread more easily (Kim et al. 2021). Furthermore, the new Omicron XBB and Omicron XBB 1.5 variants contain additional mutations in the RBD and antibody binding residues, which may further impact their dynamics and interactions with the host.

To gain deeper insights into the impact of dynamics on the binding affinity of hACE2 and antibodies with the recent Omicron XBB variants, we conducted molecular dynamics (MD) simulations. By analyzing the resulting trajectories, we investigated how these mutations influence the flexibility and rigidity of the RBD and antibody binding residues, consequently affecting their binding affinity and potential for immune evasion (**Figure 8** [↗](#)). To understand the overall flexibility changes, we measured the sum of DFI of the ACE2 binding residues, as well as the sum of DFI of the antibody binding residues, calculated from the MD trajectories and compared then with experimental viral binding (disassociation constants) and immunity evasion antibody IC50 values (Yue et al. 2023 [↗](#)).

This investigation elucidated the impact of mutations in the RBD and antibody binding residues on the binding affinity of the S protein and immune evasion by modulating their flexibility and rigidity (**Figure 8A** [↗](#)). The Omicron XBB variant exhibits heightened flexibility in hACE2 and antibody binding residues, reducing infectivity and enhancing immune evasion. Conversely, the Omicron XBB 1.5 variant induces distinct dynamics in these regions, rendering the RBD-ACE2 interface more rigid while increasing flexibility in antibody binding residues. These effects indicate that Omicron XBB 1.5 retains its antibody escape capabilities while regaining ACE2 binding affinity comparable to previous Omicron variants, in accordance with experimental findings (Yue et al. 2023 [↗](#)). These findings suggest that mutations in the RBD and antibody binding residues can have complex effects on the dynamics of the protein and, ultimately, on the virus's ability to infect and evade the host immune system through an alteration of binding site

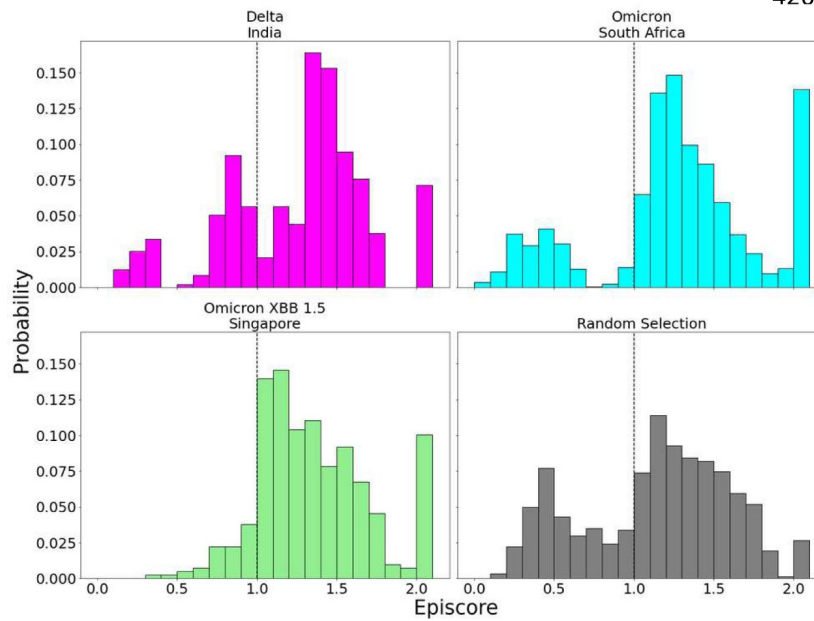


Figure 7.

EpiScores with i = low EP Delta mutation sites (magenta), low EP Omicron mutation sites (Cyan), and a random selection of sites (gray), j = site 486, and k = the binding interface of the open chain. CAP and hACE2 and antibody binding site 486 displays epistasis with almost all XBB 1.5 variant sites at almost every hACE2 binding site ($M = 1.40$, $SD = 0.46$) and presents a significantly different profile from other variant sites ($p < .01$). EpiScores involving 486 for other variants can be found in [Supplementary Figure S4](#).

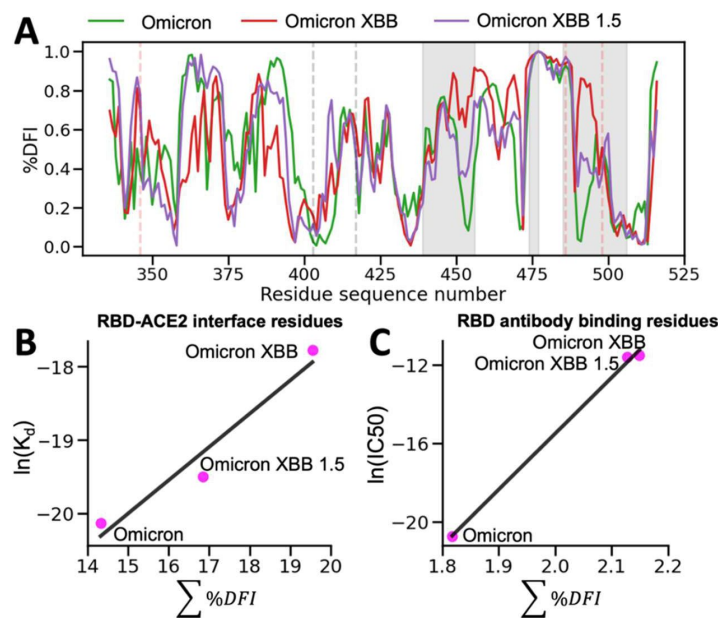


Figure 8.

The %DFI calculations for variants Omicron, XBB, and XBB 1.5. (A) %DFI profile of the variants are plotted in the same panel. The grey shaded areas and dashed lines indicate the ACE2 binding regions, whereas the red dashed lines show the antibody binding residues. (B) The sum of %DFI values of RBD-ACE2 interface residues. The trend of total %DFI with the log of K_d values overlaps with the one seen with the experiments ($R=0.97$). (C) The RBD antibody binding residues are used to calculate the sum of %DFI. The ranking captured with the total %DFI agrees with the log of IC_{50} values from the experiments.

dynamics. However, we do note that the effect of mutations will heavily depend on the genetic background in which they are mutated. The effects of a mutation on a Delta variant protein may differ entirely from the effects of a mutation on an Omicron variant protein (**Supplementary figure S5** [↗](#)).

Conclusion

We analyzed the evolutionary trajectory of the CoV-2 S protein in humans to understand the dynamic and epistatic interactions of the mutations defining specific VOCs. We first obtain the phylogenetic tree of the COV-2 S protein and identify the sites of certain recent mutations known as candidate adaptive polymorphisms (CAPs). CAPs are considered adaptive because mutations rarely tolerated in closely related sequences have suddenly become fixed, implying a degree of functional importance or evolution ([Liu et al. 2016](#) [↗](#)). In addition, our earlier work has shown that CAPs can also be compensatory; multiple CAPs may dynamically compensate for one another, changing the dynamic landscape and allowing for different mutations ([Ose, Campitelli, et al. 2022](#) [↗](#)). We then explored the mechanistic insights and epistatic relationship between the observed mutations in different VOCs and CAP sites, and, particularly, the relationship between CAP sites and the functionally critical RBD using our dynamic coupling analysis ([Kumar, A., Glembo, T.J., Ozkan, S.B. 2015b](#) [↗](#)).

We find a mechanistic pattern in the S protein evolution that is common amongst previously studied systems, where allosteric sites exert control over the dynamics of the binding sites, and mutations of these allosteric sites modulate function. Coupling our analysis with evolutionary theory showed that many of these allosteric sites regulating function of the S protein may have been subjected to adaptive evolution as observed in mutations in VOCs. Our dynamics analysis also provides a mechanistic insight where the Omicron defining sites have greater control over the binding sites than the Delta variant, and are dynamically additive with the functional advantage of CAPs, thus, the greatest infectivity may not be a coincidence.

Specifically, we find that the interactions between CAP sites and VOC-defining mutations show fingerprints of non-additive dynamics within the Delta variant. In contrast, mutations leading to the Omicron variant are largely additive, driving critical dynamical behavior closer to the patterns observed within the wild-type. These interactions may also drive observed behavior similar between the reference and Omicron strains yet differ in the delta strain, such as the severity of infection as evidenced by hospitalization rates ([Houhamdi et al. 2022](#) [↗](#); [Menni et al. 2022](#) [↗](#)). It has also been shown that the Omicron variant has a lower binding affinity with hACE2 than previous variants ([Wu et al. 2022](#) [↗](#)), which may contribute to its low pathogenicity.

Long-ranged interactions between different sites within a given protein is critically important for protein function ([Peters and Lively 1999](#) [↗](#); [Bershtein et al. 2006](#) [↗](#); [Collins et al. 2006](#) [↗](#); [Ekeberg et al. 2013](#) [↗](#); [Levy et al. 2017](#) [↗](#); [Harrigan et al. 2018](#) [↗](#); [Otten et al. 2018](#) [↗](#); [Rojas Echenique et al. 2019](#) [↗](#); [Shimagaki and Weigt 2019](#) [↗](#); [de la Paz et al. 2020](#) [↗](#); [Rizzato et al. 2020](#); [Yang et al. 2020](#) [↗](#); [Bisardi et al. 2022](#) [↗](#)) and for the CoV-2 S protein in particular ([Zeng et al. 2020](#) [↗](#); [Castiglione et al. 2021](#) [↗](#); [Dong et al. 2021](#) [↗](#); [Garvin et al. 2021](#) [↗](#); [Nielsen et al. 2022](#) [↗](#); [Ramaraio-Milne et al. 2022](#) [↗](#); [Rochman et al. 2022](#) [↗](#); [Rodriguez-Rivas et al. 2022](#) [↗](#)). By showing dynamic differences between the interactions of CAPs, which have likely played a major role in allowing the virus to infect human hosts, the binding site, and the characteristic mutations of dominant Delta and Omicron strains, we see a “fine-tuning” of protein behavior. As variants continue to evolve, Omicron sub-variants are of growing concern due in large part to further increased immune evasion ([Callaway 2022](#) [↗](#); [Wang, Guo, et al. 2022](#); [Wang, Iketani, et al. 2022](#)), and we observe that the new mutations observed in antibody binding sites yield more epistatic interaction

with the CAPs. In addition to supporting previous dynamic research on the S protein, this analysis provides the insight that CAP sites are of continued importance to protein function and should be given special attention when considering the impact of future mutations.

Methods

Dynamic Flexibility and Dynamic Coupling

The Dynamic Flexibility Index utilizes a Perturbation Response Scanning technique that combines the Elastic Network Model (ENM) and Linear Response Theory (LRT) (Gerek and Ozkan 2011 [↗](#); Nevin Gerek, Z., Kumar, S., Banu Ozkan, S. 2013). In ENM, the protein is considered as a network of beads at Cα positions interacting with each other via a harmonic spring potential. Using LRT, $\Delta\mathbf{R}$ is calculated as the fluctuation response vector of residue j due to unit force's $\mathbf{F}$ perturbation on residue i , averaged over multiple unit force directions to simulate an isotropic perturbation.

$$[\Delta\mathbf{R}]_{3N \times 1} = [\mathbf{H}]_{3N \times 3N}^{-1} [\mathbf{F}]_{3N \times 1} \quad (1)$$

$\mathbf{H}$ is the Hessian, a $3N \times 3N$ matrix that can be constructed from 3-D atomic coordinate information and is composed of the second derivatives of the harmonic potential with respect to the components of the position's vectors of length $3N$. The hessian inverse in this equation may be replaced with the covariance matrix $\mathbf{G}$ obtained from MD simulations as follows.

$$[\Delta\mathbf{R}]_{3N \times 1} = [\mathbf{G}]_{3N \times 3N} [\mathbf{F}]_{3N \times 1} \quad (2)$$

MD simulations were used to obtain the DFI profiles of Omicron, Omicron XBB, and Omicron XBB 1.5. In order to obtain DFI, each position in the structure was perturbed sequentially to generate a Perturbation Response Matrix $\mathbf{A}$

$$\mathbf{A}_{N \times N} = \begin{bmatrix} |\Delta\mathbf{R}^1|_1 & \cdots & |\Delta\mathbf{R}^N|_1 \\ \vdots & \ddots & \vdots \\ |\Delta\mathbf{R}^1|_N & \cdots & |\Delta\mathbf{R}^N|_N \end{bmatrix} \quad (3)$$

where $|\Delta\mathbf{R}^j|_i = \sqrt{\langle (\Delta\mathbf{R}^j)^2 \rangle}$ is the magnitude of fluctuation response at position i due to perturbations at position j . The DFI value of position i is then treated as the displacement response of position i relative to the net displacement response of the entire protein, which is calculated by sequentially perturbing each position in the structure.

$$\text{DFI}_i = \frac{\sum_{j=1}^N |\Delta\mathbf{R}^j|_i}{\sum_{i=1}^N \sum_{j=1}^N |\Delta\mathbf{R}^j|_i} \quad (4)$$

It is also often useful to quantify position flexibility relative to the flexibility ranges unique to individual structures. To that end, DFI can be presented as a percentile rank, %DFI. All %DFI calculations present in this work used the DFI value of every residue of the full spike structure for ranking. The DFI parameter can be considered a measure of a given amino acid position's ability to explore its local conformational space.

Dynamic Coupling Index

Similar to DFI, the dynamic coupling index (DCI) (Larrimore et al. 2017 [↗](#); Kumar, A., Glembo, T.J., Ozkan, S.B. 2015b [↗](#)) also utilizes Perturbation Response Scanning with the Elastic Network Model and Linear Response Theory. DCI captures the strength of displacement response of a given

position i upon perturbation to a single functionally important position (or subset of positions) j , relative to the average fluctuation response of position i when all of the positions within a structure are perturbed.

$$DCI_{ji} = \frac{\sum_j^{N_{\text{functional}}} |\Delta R^j|_i / N_{\text{functional}}}{\sum_{j=1}^N |\Delta R^j|_i / N} \quad (5)$$

When only positional pairs are concerned, this expression reduces to:

$$DCI_{ji} = \frac{|\Delta R^j|_i}{\sum_{j=1}^N |\Delta R^j|_i / N} \quad (6)$$

As such, this parameter represents a measure of the dynamic coupling between i and j upon a perturbation to j . As with DFI, DCI_{ji} can also be presented as a percentile-ranked % DCI_{ji} .

One of the most important aspects of DCI is that the entire network of interactions is explicitly included in subsequent calculations without the need for dimensionality reduction techniques. If one considers interactions such as communication directionality or dynamic coupling regulation between position pairs as inherent properties of an anisotropic interaction network, it is critical to include the interactions of the entire network to accurately model the effect one residue can have on another.

Here, we present two further extensions of DCI which allow us to uniquely model coupling directionality and epistatic effects: DCI_{asym} and EpiScore, respectively. Interestingly, we can capture asymmetry between different residues within a protein through DCI, as a coupling in and of itself is asymmetric within an anisotropic network. That is, each amino acid has a set of positions to which it is highly coupled, and this anisotropy in connections gives rise to unique differences in coupling between a given i j pair of amino acids which do not have direct interactions (**Figure 2A**). DCI_{asym} , then, is simply DCI_{ij} (the normalized displacement response of position j upon a perturbation to position i) – DCI_{ji} (**Equation (7)**). Using DCI_{asym} we can determine a cause-effect relationship between the i j pair in terms of force/signal propagation between these two positions.

$$DCI_{\text{asym}} = DCI_{ij} - DCI_{ji} \quad (7)$$

$$\%DCI_{\text{asym}} = \%DCI_{ij} - \%DCI_{ji} \quad (8)$$

where a positive DCI_{asym} value indicates communication from position i to position j .

EpiScore can identify or describe potential non-additivity in substitution behavior between residue pairs. This metric can capture the differences in a normalized perturbation response to a position k when a force is applied at two residues i and j simultaneously versus the average additive perturbation response when each residue i , j , is perturbed individually (**Figure 5A**, **Equation 9**).

$$\text{EpiScore} = \frac{\%DCI_{\{ij\}k}}{\frac{1}{2}(\%DCI_{ik} + \%DCI_{jk})} \quad (9)$$

EpiScore values < 1 (> 1) indicate that the additive perturbations of positions i and j generates a greater (lesser) response at position k than the effect of a simultaneous perturbation. This means that, when treated with a simultaneous perturbation at both sites i and j , the displacement response of k is lower (higher) than the average effect of individual perturbations to i and j , one at a time. As EpiScore is a linear scale, the further the value from 1, the greater the effect described above.

Molecular Dynamics (MD)

The production simulations for the Omicron variants, including Omicron, Omicron XBB, and Omicron XBB 1.5, were performed with the AMBER software package. These variants, each characterized by specific mutations, were modeled based on the template PDB structure 6M0J. The initial protein configurations in the simulations were parameterized using the ff14SB force field (Maier et al. 2015 [↗](#)). In order to create an appropriate solvation environment for the proteins, a solvation box was defined around them, maintaining a minimum separation distance of 16Å from the protein to the box boundaries. This was accomplished by employing the explicit TIP3P water model (Sun 1995), with the addition of sodium and chloride ions to maintain overall charge neutrality.

The simulation procedure involves an initial energy minimization step, aimed at mitigating steric clashes and optimizing the system's energy. A steepest descent algorithm was applied, encompassing 11,000 steps. Subsequently, the system underwent a gradual temperature increase (heat up), up to 300K, and was subjected to production simulations under a constant number of particles, pressure, and temperature ensemble (NPT).

During these production simulations, temperature was maintained at 300K, with pressure regulation set at 1 bar. Temperature regulation was achieved through the utilization of the Langevin thermostat (Hünenberger 2005) and Berendsen barostat (Berendsen et al. 1984 [↗](#)), featuring a collision frequency of 1.0 picoseconds⁻¹. Hydrogen atom bond lengths were constrained using the SHAKE algorithm (Pearlman et al. 1995 [↗](#)). The production trajectories were simulated for 1µs each.

To ensure the reliability of the simulations and assess their convergence, a convergence criterion was employed. The achieved convergence was determined by monitoring the root mean square deviation (RMSD) between the highest sampled conformation in consecutive time windows (Sawle and Ghosh 2016 [↗](#)). Specifically, convergence was defined as the point at which the RMSD between the highest sampled conformation in the last 300 nanoseconds (ns) window and the 300ns window immediately preceding it dropped below 1 angstrom (Å). Window sizes varying from 100ns to 500ns were employed to evaluate convergence, ensuring the robustness and stability of the obtained results.

To calculate DFI, covariance matrix data were computed over different time windows as discussed above. By default, utilizing the Hessian implies a restriction to a harmonic potential, assuming that the data are sampled from a Gaussian distribution. Ergodicity in both simulation time and initial structures sampled in each time interval ensures two key conditions: (i) Consistency of potential energy across conformations sampled from the same distribution. (ii) The sampling of different initial conformations while computing covariance matrices at various time windows eliminates global motions and accurately captures equilibrium coordinates. Consequently, the final average DFI profiles are independent of time window size, resulting in consistent results across different time window sizes (e.g., 50 ns vs. 75 ns) and enabling the acquisition of statistically significant DFI values.

Data and Resource Availability

The code to perform DFI and DCI analysis is available at <https://github.com/SBOZKAN/DFI-DCI>.

Molecular Dynamics data are available upon request. The mutation sites and EP values are contained in the supporting information files as “Supplementary_mutation_info.csv”. The alignment used to generate EP values is also contained within the supporting information files as “EP_alignment.fas”.

Protein Databank ID number 6VXX (Walls et al. 2020) was used for closed conformation DCI calculations. 6VSB (Wrapp et al. 2020) was used for DFI calculations, EpiScore calculations, and open conformation DCI calculations. 6M0J (Lan et al. 2020) was used in molecular dynamics simulations of the RBD.

Acknowledgements

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Supplemental Figures

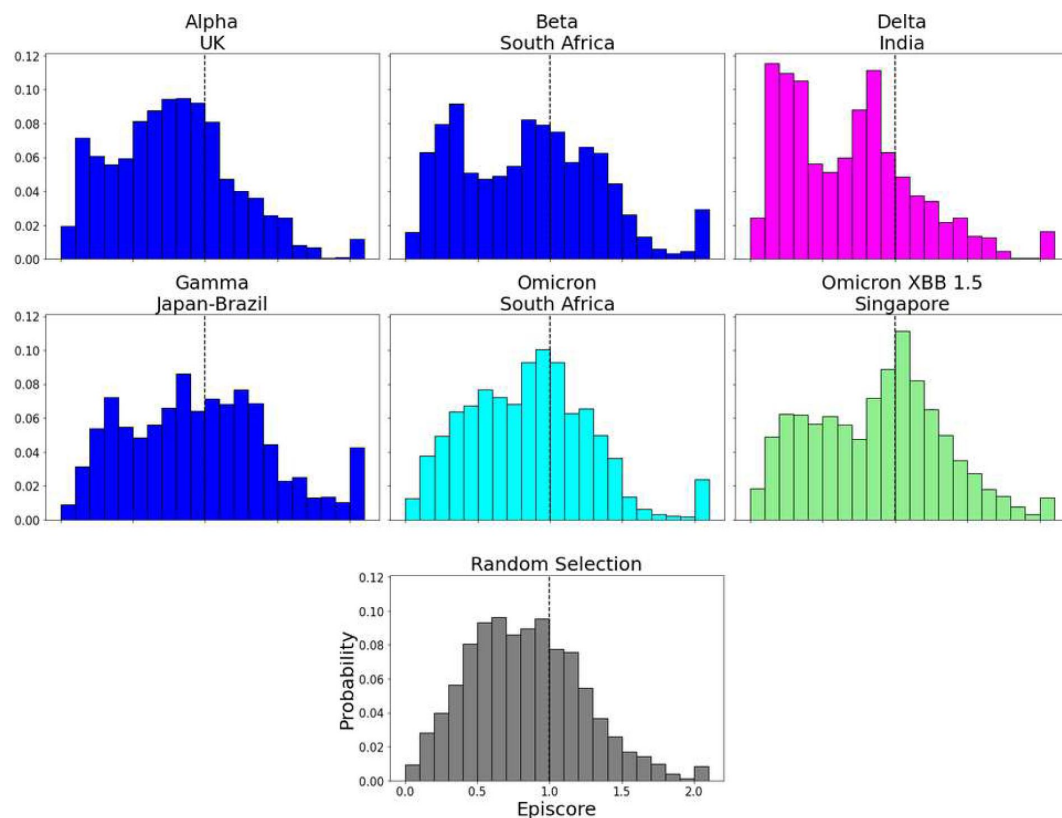
	18	19	20	26	27	67	80	83	95	138	142	145	146	158	183	190	212	213	215	252	339	346	368	371	373	375	376	405	408	417	440	445
Wild-type (MN90894.3)	L	T	T	P	A	A	D	V	T	D	G	Y	H	R	Q	R	L	V	D	G	G	R	L	S	S	S	T	D	R	K	N	V
Alpha	L	T	T	P	A	A	D	V	T	D	G	Y	H	R	Q	R	L	V	D	G	G	R	L	S	S	S	T	D	R	K	N	V
Beta	L	T	T	P	A	A	A	V	T	D	G	Y	H	R	Q	R	L	V	D	G	G	R	L	S	S	S	T	D	R	N	N	V
Gamma	F	T	N	S	A	A	D	V	T	Y	G	Y	H	R	Q	S	L	V	D	G	G	R	L	S	S	S	T	D	R	T	N	V
Delta	L	R	T	P	A	A	D	V	T	D	G	Y	H	G	Q	R	L	V	D	G	G	R	L	S	S	S	T	D	R	K	N	V
Omicron	L	T	T	P	A	V	D	V	I	D	G	D	H	R	Q	R	I	V	D	G	D	R	L	L	P	F	T	D	R	N	K	V
Omicron XBB 1.5	L	I	T	P	S	A	D	A	T	D	D	Y	Q	R	E	R	L	E	D	V	H	T	I	F	P	F	A	N	S	N	K	P

	446	452	460	477	478	484	486	490	493	496	498	501	505	547	570	614	655	679	681	701	716	764	796	856	950	954	969	981	982	1027	1118	1176
Wild-type (MN90894.3)	G	L	N	S	T	E	F	F	Q	G	Q	N	Y	T	A	D	H	N	P	A	T	N	D	N	D	Q	N	L	S	T	D	V
Alpha	G	L	N	S	T	E	F	F	Q	G	Q	Y	Y	T	D	G	H	N	H	A	I	N	D	N	D	Q	N	L	A	T	H	V
Beta	G	L	N	S	T	K	F	F	Q	G	Q	Y	Y	T	A	G	H	N	P	V	T	N	D	N	D	Q	N	L	S	T	D	V
Gamma	G	L	N	S	T	K	F	F	Q	G	Q	Y	Y	T	A	G	Y	N	P	A	T	N	D	N	D	Q	N	L	S	I	D	F
Delta	G	R	N	S	K	E	F	F	Q	G	Q	N	Y	T	A	G	H	N	R	A	T	N	D	N	N	Q	N	L	S	T	D	V
Omicron	S	L	N	N	K	A	F	F	R	S	R	Y	H	K	A	G	Y	K	H	A	T	K	Y	K	D	H	K	F	S	T	D	V
Omicron XBB 1.5	S	L	K	N	K	A	P	S	Q	G	R	Y	H	T	A	G	Y	K	H	A	T	K	Y	N	D	H	K	L	S	T	D	V

Supplementary figure S1.

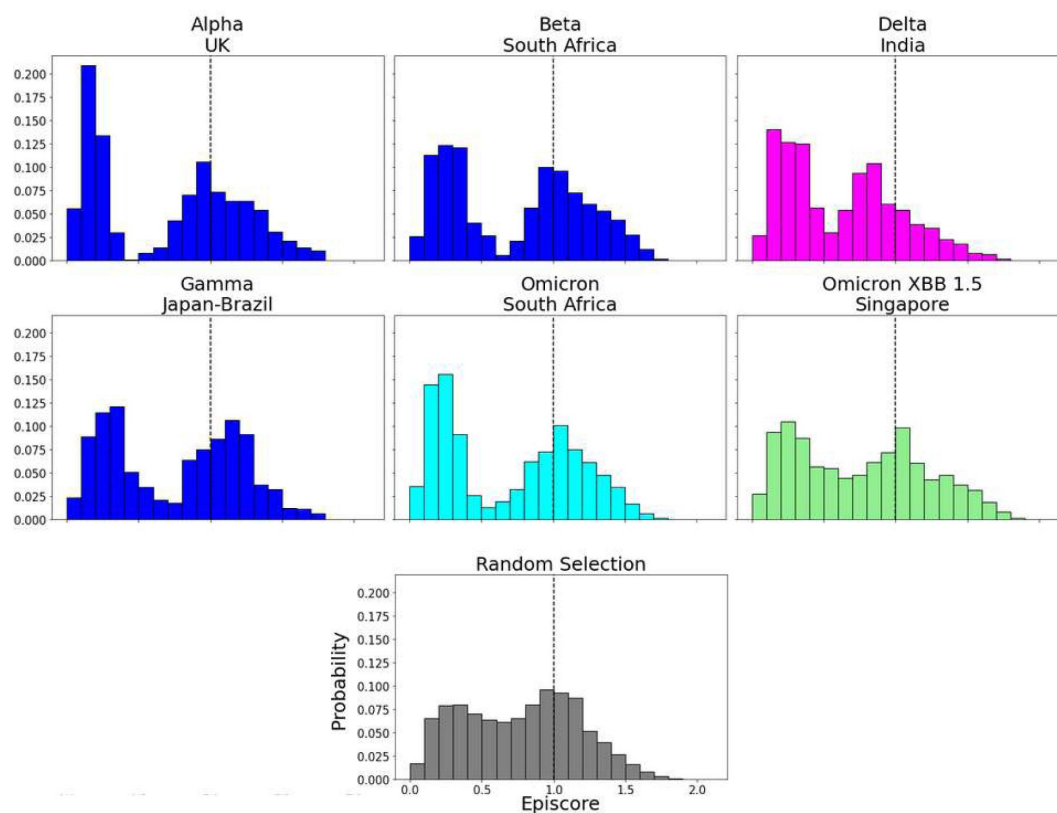
A comparison of the amino acid sequences for different variant spike proteins used in this paper

Sequence MN90894.3 is used as the wild-type (reference) sequence. For each different variant, residue positions matching the reference sequence are shown in blue, while locations where the residue differs from the reference sequence are shown in yellow. For S protein locations not shown here, all variants presented have the same amino acid type as the reference sequence.



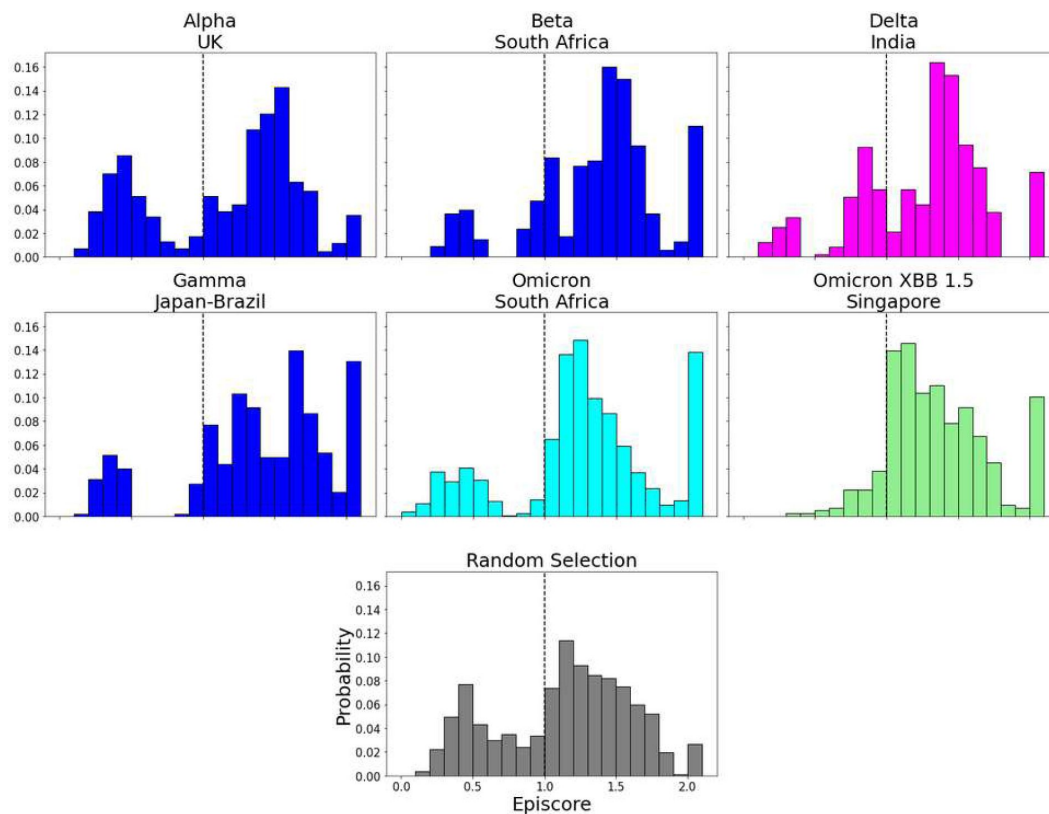
Supplementary figure S2.

EpiScores with i = characteristic mutation sites, j = low EP sites, and k = the binding interface of the open chain. EpiScores using variant sites are significantly different ($p < .001$) from a set of EpiScores using random sites.



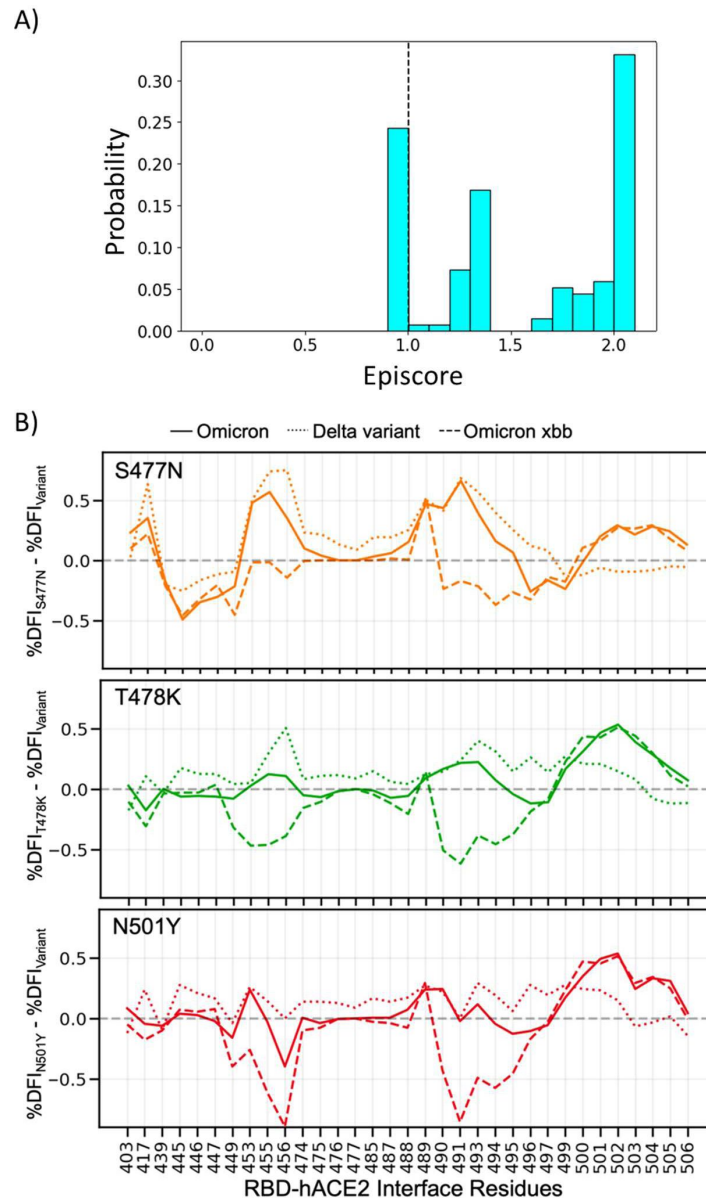
Supplementary figure S3.

EpiScores with i = characteristic mutation sites within the NTD, j = low EP sites, and k = the binding interface of the open chain. NTD domain mutation sites result in markedly lower EpiScores compared to elsewhere.



Supplementary figure S4.

EpiScores with i = characteristic mutation sites, j = site 486, and k = the binding interface of the open chain. CAP and hACE2 and antibody binding site 486 displays epistasis with almost all XBB 1.5 variant sites at almost every hACE2 binding site. EpiScores using variant sites are significantly different ($p < .001$) from a set of EpiScores using random sites.



Supplementary figure S5.

(A) Three critical mutations that emerged quickly, and were frequently observed in other dominant variants are S477N, T478K, and N501Y. EpiScores of sites 477, 478, and 501 with one another are shown (with k = the binding interface of the open chain). These residues are highly epistatic, producing higher responses than expected when perturbed together. (B) The difference in the dynamic flexibility profiles between the single mutants and the most common variants for the hACE2 binding residues of the RBD. The DFI profiles exhibit significantly different flexibility in each background variant, highlighting the critical non-additive interactions of the other mutation in the given background variant. Thus, these three binding affinity impacting mutations do not contribute only their own effects to the binding interface, there are epistatic interactions with the other mutations in VOCs that shape the dynamics of the binding interface to modulate binding.

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Editors

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Reviewer #2 (Public Review):

The authors set out to identify CAPs (Candidate Adaptive Polymorphisms), i.e., simply put mutations that carry a potential functional advantage, and utilize computational methods based on the perturbation of C-alpha positions with an Elastic Network Model to determine if dynamics of CAP residues are different in any way.

The authors have addressed the main methodological concerns.

However, one point remains. The specific comparison of which CAPs have been previously identified by other means, particularly with other computational methods that look into dynamics is still lacking. It is unfortunate that the authors do not present such analysis, particularly with respect to single point mutational analysis of Teruel et al. in Plos Comp. Bio. If CAP positions were previously identified by other means it adds strength to the methodology used by the authors. The authors also do not discuss their results in light of the work of Lam et al. (Sci. Comm, 2020) where an evolutionary analysis of Spike/ACE2 binding across homologues is performed. I believe that such deeper discussion of the current results in light of previous work, adds strength to the analysis presented in this manuscript as the methodology is different. Even if all results were not new, with the method being different from the other means by which such results were obtained, it would be still a worthy contribution to the field. Furthermore, for the community at large trying to understand the importance of particular positions in Spike, knowing that a particular position identified here was also identified by works X, Y, Z adds a lot of to the field. I can only think that the authors may imagine that if one of their CAPs was identified by other means previously, it takes away from the merit of their work, but it is actually the opposite. I urge the authors to not brush away this. In fact, more important than any methodological aspect of the present work, this strengthening of evidence for particular positions by several independent methods is the most important evidence that the authors can contribute to the field.

<https://doi.org/10.7554/eLife.92063.2.sa1>

Reviewer #3 (Public Review):

Summary:

The manuscript uses a combination of evolutionary approaches and structural/dynamics observations to provide mechanistic insights in the adaptation of the Spike protein during the evolution of SARS-COV-2 variants. The conclusion that CAP sites should be taken in particular account when considering the impact of the emergence of new strains and mutations is warranted.

Strengths:

The results presented in this work are very well outlined with well-written text, pleasant and well-described pictures, didactical and clear description of the methods, e.g. the discussion of how the MD equilibration procedure is applied and evaluated is clear and well argument. The citation of relevant similar results with different approaches strengthens the reasoning; in particular, comparing the calculated scores with previous experimentally obtained data is one of the strongest points of the manuscript.

Weaknesses:

There are no replicas of the molecular dynamics (MD) simulations, understandable since it's not a MD-focused paper. However, the comparison of multiple replicas could enhance the reliability of the findings.

Author Response

The following is the authors' response to the original reviews.

We would like to extend our gratitude to the reviewers for their meticulous analysis and constructive feedback on our manuscript. We have revised our paper based on the suggestions regarding supporting literature and the theory behind CAPs along with detailed insights regarding our methods. Their suggestions have been extremely useful in strengthening the clarity and rigor of our manuscript.

Reviewer #1 (Recommendations For The Authors):

(1) There are no obvious problems with this paper and it is relatively straightforward. There are some challenges that I would like to suggest. These variants have multiple mutations, so it would be interesting if you could drill down to find out which mutation is the most important for the collective changes reported here. I would like to see a sequence alignment of these variants, perhaps in the supplemental material, just to get some indication of the extent of mutations involved.

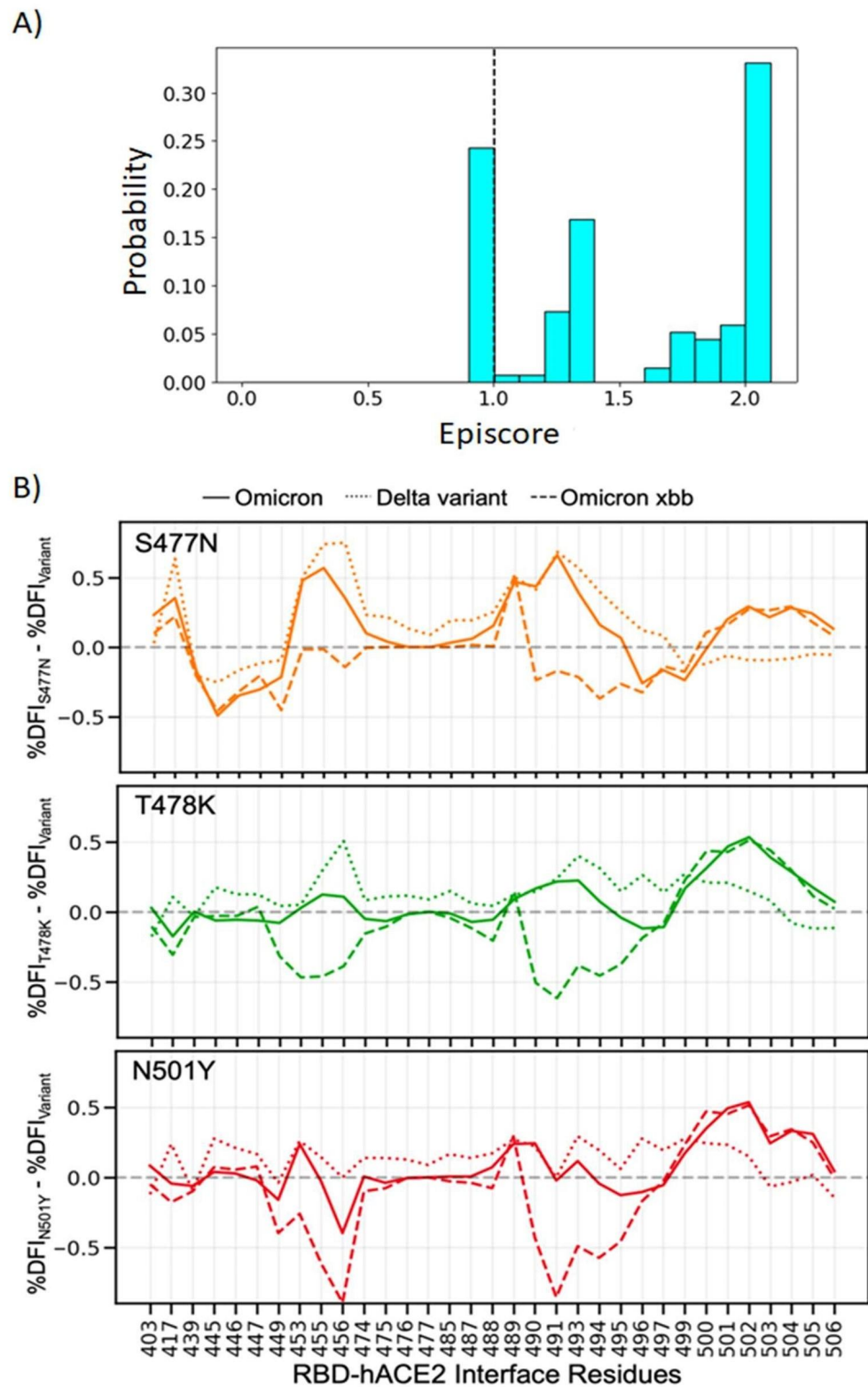
Finding the most important mutation within a set is a tricky question, as each mutation changes the way future mutations will affect function due to epistasis. Indeed, this is what we aim to explore in this work. To illustrate this point, we included a new supplementary figure S5A. Three critical mutations that emerged quickly, and were frequently observed in other dominant variants, were S477N, T478K, and N501Y. Thus, we computed the EpiScore values of these three mutations, with several critical residues contributing to hACE2 binding. The EpiScore distribution indicates that residues 477, 478, and 501 have strong epistatic (i.e., non-additive) interactions, as indicated by EpiScore values above 2.0.

To further investigate these epistatic interactions, we first conducted MD simulations and computed the DFI profile of these three single mutants. We analyzed how different the DFI scores of the hACE2 binding interface residues of the RBD are, across three single mutants with Omicron, Delta, and Omicron XBB variants (Fig S5B). Fig S5B shows how mutations at these particular sites affect the binding interface DFI in various backgrounds, as the three mutations are also observed in the Omicron, XBB, and XBB 1.5 variants. If the difference in the DFI profile of the mutant and the given variant is close to 0, then we could safely state that this mutation affected the variant the most. However, what we observe is quite the opposite: the DFI profile of the mutation is significantly different in different variant backgrounds. While these mutations may change overall behavior, their individual contributions to overall function are more difficult to pin down because overall function is dependent on the non-additive interactions between many different residues.

Author response image 1.

(A) Three critical mutations that emerged quickly, and were frequently observed in other dominant variants, were S477N, T478K, and N501Y. EpiScores of sites 477, 478, and 501 with one another are shown with k = the binding interface of the open chain. These residues are highly epistatic, producing higher responses than expected when perturbed together. (B) The difference in the dynamic flexibility profiles between the single mutants and the most common variants for the hACE2 binding residues of the RBD. DFI profiles exhibit significant variation from zero, and also show different flexibility in each background variant, highlighting the critical non-additive interactions of the other mutation in the given background variant. Thus, these three critical mutations, impacting binding affinity, do not solely contribute to the binding. There are epistatic interactions with the other mutations in

VOCs that shape the dynamics of the binding interface to modulate binding affinity with hACE2.



As we discussed above, while the epistatic interactions are crucial and the collective impact of the mutations shape the mutational landscape of the spike protein, we would like note that

mutation S486P is one of the critical mutations we identify, modulating both antibody and hACE2 binding and our analysis reveals the strong non-additive interactions with the other mutational sites. This mutational site appears in both XBB1.5 and earlier Omicron strains which highlights its importance in functional evolution of the spike protein. CAPs 346R, 486F, and 498Q also may be important, as they have a high EpiScore, indicating critical epistatic interaction with many mutation sites.

Regarding to the suggestion about presenting the alignment of the different variants, we have attached a mutation table, highlighting the mutated residues for each strain compared to the reference sequence as supplemental Figure S1 along with the full alignment file.

(2) Also, I am wondering if it would be possible to insert some of these flexibilities and their correlations directly into the elastic network models to enable a simpler interpretation of these results. I realize this is beyond the scope of the present work, but such an effort might help in understanding these relatively complex effects.

This is great suggestion. A similar analysis has been performed for different proteins by Mcleash (See doi: 10.1016/j.bpj.2015.08.009) by modulating the spring constants of specific position to alter specific flexibility and evaluate change in elastic free energy to identify critical mutation (in particular, allosteric mutation) sites. We will be happy to pursue this as future work.

Minor

(3) 1 typo on line 443 - should be binding instead of biding.

Fixed, thanks for spotting that.

(4) The two shades of blue in Fig. 4B were not distinguishable in my version.

To fix this, we have changed the overlapping residues between Delta and Omicron to a higher contrast shade of blue.

(5) Compensatory is often used in an entirely different way - additional mutations that help to recover native function in the presence of a deleterious mutation.

Although our previous study (Ose et al. 2022, Biophysical Journal) shows that compensatory mutations were generally additive, the two ideas are not one and the same. We thank the reviewer for pointing this out. Therefore, to clarify, we have now described our results in terms of dynamic additivity, rather than compensation.

Reviewer #2 (Recommendations For The Authors):

(1) The authors note that the identified CAPs overlap with those of others (Cagliani et al. 2020; Singh and Yi 2021; Starr, Zepeda, et al. 2022). In itself, this merits a deeper discussion and explicit indication of which positions are not identified. However, there is one point that I believe may represent a fundamental flaw in this study in that the calculation of EP from the alignment of S proteins ignores entirely the differences in the interacting interface with which S for different coronaviruses in the alignment interact in the different receptors in each host species. This may be the reason why so many "CAPs" are in the RBD. The authors should at the very least make a convincing case of why they are not simply detecting constraints imposed by the different interacting partners, at least in the case of positions within the RBD interface with ACE2. Another point that the authors should discuss is that ACE2 is not the only receptor that facilitates infection,

TMPRSS2 and possibly others have been identified as well. The results should be discussed in light of this.

To begin with, we have now explicitly noted (on line 135) that “sites 478, 486, 498, and 681 have already been implicated in SARS-CoV-2 evolution, leaving the remaining 11 CAPs as undiscovered candidate sites for adaptation.” Evolutionary analyses are done using orthologous protein sequences, so there is no way to integrate information on different receptors in each host species in the calculation of EPs. However, we appreciate that the preponderance of CAPs in the RBD is likely due to different binding environments. We have added the following text (on line 83) to clarify our point: “Adaptation in this case means a virus which can successfully infect human hosts. As CAPs are unexpected polymorphisms under neutral theory, their existence implies a non-neutral effect. This can come in the form of functional changes (Liu et al. 2016) or compensation for functional changes (Ose et al. 2022). Therefore, we suspect that these CAPs, being unexpected changes from coronaviruses across other host species with different binding substrates, may be partially responsible for the functional change of allowing human infection.” This hypothesis is supported by the overlap of CAPs we identified with the positions identified in other studies (e.g., 478, 486, 498, and 681). Binding to TMPRSS2 and other substrates are also covered by this analysis as it is a measure of overall evolutionary fitness, rather than binding to any specific substrate. Our paper does focus on discussing hACE2 binding and mentions furin cleavage, but indeed lacks discussion on the role of TMPRSS2. We have added the following text to line 157: “Another host cell protease, TMPRSS2, facilitates viral attachment to the surface of target cells upon binding either to sites Arg815/Ser816, or Arg685/Ser686 which overlaps with the furin cleavage site 676-689, further emphasizing the importance of this area (Hoffmann et al. 2020b; Fraser et al. 2022).”

(2) Turning now to the computational methods utilized to study dynamics, I have serious reservations about the novelty of the results as well as the validity of the methodology. First of all, the authors mention the work of Teruel et al. (PLOS Comp Bio 2021) in an extremely superficial fashion and do not mention at all a second manuscript by Teruel et al. (Biorxiv 2021.12.14.472622 (2021)). However, the work by Teruel et al. identifies positions and specific mutations that affect the dynamics of S and the evolution of the SARS-CoV-2 virus in light of immune escape, ACE2 binding, and open and closed state dynamics. The specific differences in approach should be noted but the results specifically should be compared. This omission is evident throughout the manuscript. Several other groups have also published on the use of normal-mode analysis methods to understand the Spike protein, among them Verkhivker et al., Zhou et al., Majumder et al., etc.

Thank you for your suggestions. Upon further examination of the listed papers, we have added citations to other groups employing similar methods. However, it's worth noting that the results of Teruel et al.'s studies are generally not directly comparable to our own. Particularly, they examine specific individual mutations and overall dynamical signatures associated with them, whereas our results are always considered in the context of epistasis and joint effects with CAPs, and all mutations belong to the common variants. Although important mutations may be highlighted in both cases, it is for very different reasons. Nevertheless, we provide a more detailed mention of the results of both studies. See lines 178, 255, and 393.

(3) The last concern that I have is with respect to the methodology. The dynamic couplings and the derived index (DCI) are entirely based on the use of the elastic network model presented which is strictly sequence-agnostic. Only C-alpha positions are taken into consideration and no information about the side-chain is considered in any manner. Of course, the specific sequence of a protein will affect the unique placement of C-alpha

atoms (i.e., mutations affect structure), therefore even ANM or ENM can to some extent predict the effect of mutations in as much as these have an effect on the structure, either experimentally determined or correctly and even incorrectly modelled. However, such an approach needs to be discussed in far deeper detail when it comes to positions on the surface of a protein such that the reader can gauge if the observed effects are the result of modelling errors.

We would like to clarify that most of our results do not involve simulations of different variants, but rather how characteristic mutation sites for those variants contribute to overall dynamics. For the full spike, we operate on only two simulations: open and closed. When we do analyze different variants, starting on line 438, the observed difference does not come from the structure, but from the covariance matrix obtained from molecular dynamics (MD) simulations, which are sensitive to single amino acid changes.

Reviewer #3 (Recommendations For The Authors):

(1) On line 99 there is a misspelling, 'withing'.

It has been fixed. Thanks for spotting that.

(2) Some graphical suggestions to make the figures easier to read:

In Figure 1C, a labeled circle around the important sites, the receptor binding domain, and the Furin cleavage site, would help the reader orient themselves. Moreover, it would make clear which CAPs are NOT in the noteworthy sites described in the text.

Good idea. We have added transparent spheres and labels to show hACE2 binding sites and Furin cleavage sites.

In Figure 2C the colors are a bit low contrast; moreover, there are multiple text sizes on the same figure which should perhaps be avoided to ensure legibility.

We have made yellow brighter and standardized font sizes.

Figure 3 is a bit dry, perhaps indicating in which bins the 'interesting' sites could be informative.

Thank you for the suggestion, but the overall goal of Figure 3 is to illustrate that the mutational landscape is governed by the equilibrium dynamics in which flexible sites undergo more mutations during the evolution of the CoV2 spike protein. Therefore, adding additional positional information may complicate our message.

Figure 4, the previous suggestions about readability apply.

We ensured same sized text and higher contrast colors.

Figure 5B, the residue labels are too small.

We increased the font size of the residue labels.

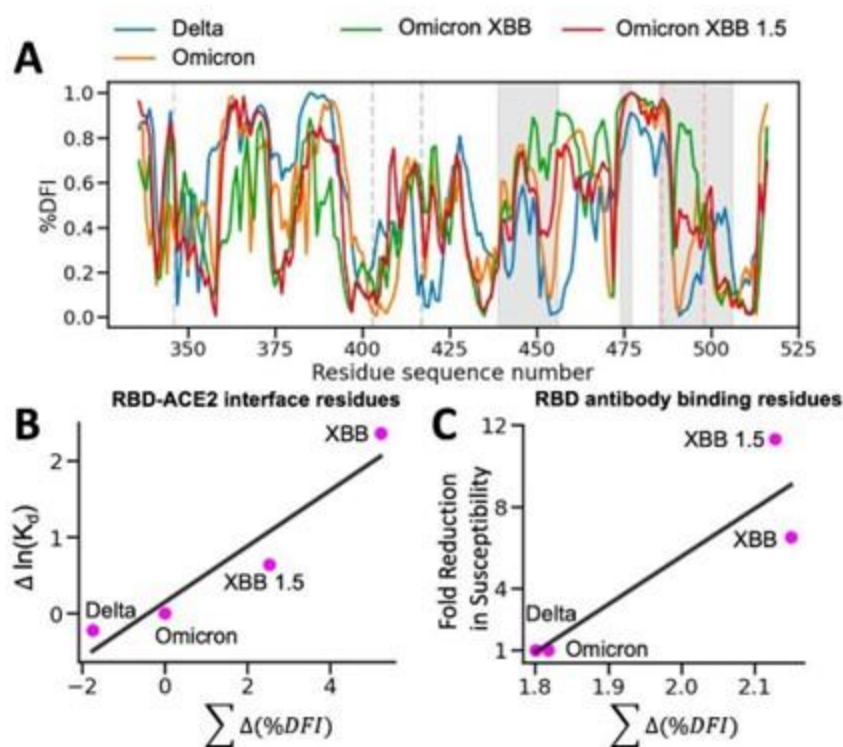
In Figure 8 maybe adding Delta to let the reader orient themselves would be helpful to the discussion.

Unfortunately, there is no single work that has experimentally quantified binding affinities towards hACE2 for all the variants. When we conducted the same analysis for the Delta

variant in Figure 8, the experimental values were obtained from a different source (doi: 10.1016/j.cell.2022.01.001) and the values were significantly different from the experimental work we used for Omicron (Yue et al. 2023). When we could adjust based on the difference in experimentally measured binding affinity values of the original Wuhan strain in these two separate studies, we observed a similar correlation, as seen below. However, we think this might not be a proper representation. Therefore, we chose to keep the original figure.

Author response image 2.

The %DFI calculations for variants Delta, Omicron, XBB, and XBB 1.5. (A) %DFI profile of the variants are plotted in the same panel. The grey shaded areas and dashed lines indicate the ACE2 binding regions, whereas the red dashed lines show the antibody binding residues. (B) The sum of %DFI values of RBD-hACE2 interface residues. The trend of total %DFI with the log of Kd values overlaps with the one seen with the experiments. (C) The RBD antibody binding residues are used to calculate the sum of %DFI. The ranking captured with the total %DFI agrees with the susceptibility fold reduction values from the experiments.



(3) Replicas of the MD simulations would make the conclusions stronger in my opinion.

We ran a 1 μ s long simulation and performed convergence analysis for the MD simulations using the prior work (Sawle L, Ghosh K. 2016.) More importantly, we also evaluated the statistical significance of computed DFI values as explained in detail below (Please see the answer to question 3 of Reviewer #3 (Public Review):)

Reviewer #3 (Public Review):

(1) A longer discussion of how the 19 orthologous coronavirus sequences were chosen would be helpful, as the rest of the paper hinges on this initial choice.

The following explanation has been added on line 114: EP scores of the amino acid variants of the S protein were obtained using a Maximum Likelihood phylogeny (Kumar et al. 2018) built from 19 orthologous coronavirus sequences. Sequences were selected by examining available non-human sequences with a sequence identity of 70% or above to the human SARS CoV-2's S protein sequence. This cutoff allows for divergence over evolutionary history such that each amino acid position had ample time to experience purifying selection, whilst limiting ourselves to closely related coronaviruses. (Figure 1A).

(2) The 'reasonable similarity' with previously published data is not well defined, nor there was any comment about some of the residues analyzed (namely 417-484). We have revised this part of the manuscript and add to the revised version.

We removed the line about reasonable similarity as it was vague, added a line about residues 417-484, and revised the text accordingly, starting on line 354.

(3) There seem to be no replicas of the MD simulations, nor a discussion of the convergence of these simulations. A more detailed description of the equilibration and production schemes used in MD would be helpful. Moreover, there is no discussion of how the equilibration procedure is evaluated, in particular for non-experts this would be helpful in judging the reliability of the procedure.

We opted for a single, extended equilibrium simulation to comprehensively explore the longterm behavior of the system. Given the specific nature of our investigation and resource constraints, a well-converged, prolonged simulation was deemed a practical and scientifically valid approach, providing a thorough understanding of the system's dynamics. (doi: 10.33011/livecoms.1.1.5957, <https://doi.org/10.1146/annurev-biophys-042910-155255>)

We updated our methods section starting on line 605 with extended information about the MD simulations and the converge criteria for the equilibrium simulations. We also added a section that explains our analysis to check statistical significance of obtained DFI values.